EARSAFE METER V1

Battery-powered desk sound monitor for personal sound exposure control.

DOCUMENT	EM-V1
REVISION	A
STATUS	INITIAL ISSUE
OWNER	BUILDERFRIDAY
EXPORT	DIRECT PDF

Design intent

- Measure room sound near the listener position.
- Show one clear sound level value at a glance.
- Show warning state before daily sound dose becomes excessive.
- Run from internal battery power with USB-C charge.

CONCEPT VIEW

Desk instrument enclosure, front view

SECTION 00

Contents

This specification defines the first build of EARSAFE METER. The device is a battery-powered desk sound monitor. It shows present sound level, warning state, peak level, and daily sound dose.

DOCUMENT INTENT

This manual must stay direct. It must stay technical. It must not use web-document decoration, startup language, or sales language.

SECTION 01

Purpose and Use Limits

EARSAFE METER V1 is a desk sound monitor. The device measures sound near the user position. The device shows present A-weighted sound level. The device also shows warning state, peak level, and daily sound dose.

Use the device for personal sound control during music use, speaker testing, long study sessions, and other repeated listening activity. Do not use the device as a certified legal meter. Do not use the device as the only control for industrial safety.

USE LIMIT

The reading becomes useful only after calibration against a known reference meter. Before calibration, use the value as relative guidance.

SECTION 02

Build of Record

ITEM	FINAL PICK	WHY THIS IS THE PICK
Main board	Seeed Studio XIAO nRF52840 Sense	USB-C, onboard PDM microphone, onboard battery charger, very small board, official support in Arduino and CircuitPython.
Display	Waveshare 2.13 inch e-Paper HAT, black and white, 250 by 122	Paper-like display, low power, compact board, easy SPI wiring.
Battery	Adafruit 3.7 V 2000 mAh Li-ion polymer battery, product 2011	Known JST-PH battery, protected, compact, enough capacity for a desk device.
Display alert	Display only	No buzzer in V1. Keep the build simpler and quieter.

Wireless	Disabled in V1	Do not spend V1 effort on phone setup or background radio power use.
Core outputs	Current dBA, one-minute average, peak, daily dose	These values are enough to guide listening behavior.

- Keep the first build simple.
- Use one exact part for each function.
- Do not start case design before the bench tests pass.
- Do not add Wi-Fi, buzzer, or a custom PCB in V1.

V1 BOUNDARY

This document fixes the V1 design. If a part is not listed here, do not add it to the first build. The goal is a working desk instrument, not maximum features.

SECTION 03

Test-First Build Strategy

Build the device in gates. Do not move to the next gate until the current gate passes. This is the main beginner rule for the project.

GATE	OBJECTIVE	PASS CONDITION
G-01	Power the XIAO board from USB-C and then from the LiPo battery.	Board boots in both cases. Charge LED works with battery connected.
G-02	Wire the e-paper display and show a static test screen.	The display refreshes fully and shows the expected image.
G-03	Read microphone samples and print raw level data.	Microphone values change when room sound changes.
G-04	Calculate and display current dBA, average, and peak.	The values update in a stable and believable way.
G-05	Run calibration against a reference meter.	Stored offset makes the display agree with the reference close enough for personal use.
G-06	Only after the above: design and print the enclosure.	The finished hardware still passes Gates 01 to 05 after assembly.

STOP RULE

If Gate 02 fails, do not write sound-processing code. If Gate 03 fails, do not design the enclosure. If Gate 05 fails, do not trust the displayed dBA value.

SECTION 04

Hardware Configuration

This section is the exact parts list for the first build. It is the build of record.

LINE	PART	EXACT PICK	WHAT IT DOES
1	Main controller	Seeed Studio XIAO nRF52840 Sense	Runs firmware, provides USB-C, provides the onboard microphone, and charges the battery.
2	Display	Waveshare 2.13 inch e-Paper HAT, black and white	Shows the large dBA value and all warning states.
3	Battery	Adafruit 3.7 V 2000 mAh Li-ion polymer battery, product 2011	Provides portable power for V1.
4	Hookup wire	Eight short 28 AWG stranded wires	Connects the XIAO board to the e-paper board.
5	Solder	Lead-free rosin-core electronic solder	Required for permanent wiring.
6	Buttons	None for first bench build	Use firmware defaults first. Add buttons only after the display and microphone both work.
7	Alert hardware	None in V1	Display state is the only alert output in V1.

ITEM	DIMENSIONS	QTY	NOTES
XIAO board	21 by 17.8 mm	1	Small enough to mount behind the display.
Waveshare display board	65.0 by 30.2 mm	1	This is the board size, not only the visible screen.
Visible display area	48.55 by 23.71 mm	1	Use this for the front window cutout.
Battery	60 by 36 by 7 mm	1	Use this size for the first enclosure volume check.

> BEGINNER ADVICE
>
> Buy the exact parts before you design the enclosure. Do not design the case around a guessed battery size or a guessed display board size.

SECTION 05

Wiring and Electrical Design

The sound enters through the digital microphone. The controller reads the signal. The firmware applies A-weighting and RMS logic. The firmware then computes present dBA, one minute average, peak, and daily dose.

MIC → MCU INPUT → A-WEIGHT / RMS → DISPLAY

USB-C / CELL → POWER CONTROL → SETTINGS / OFFSET → ALERT

Figure 1. The upper line shows the signal chain. The lower line shows the power and settings chain.

DISPLAY PIN	CONNECT TO	XIAO PIN	COMMENT
VCC	3V3	3V3	Power the display from 3.3 V.
GND	Ground	GND	Common ground is mandatory.
DIN	SPI MOSI	D10	XIAO SPI data out.
CLK	SPI SCK	D8	XIAO SPI clock.
CS	GPIO	D3	Chip select for the display.
DC	GPIO	D2	Data or command select.
RST	GPIO	D1	Display hardware reset.
BUSY	GPIO input	D0	Display ready or busy feedback.

WIRING RULE

Do not connect display VCC to 5 V. Do not use MISO. Keep the display wires short. For the first bench build, target wire lengths of about 60 to 90 mm.

SOFTWARE ITEM	FINAL PICK	WHY
IDE	Arduino IDE 2.x	Simpler beginner path than a custom toolchain.
Board package	Seeed nRF52 board support for XIAO nRF52840 Sense	Official board support path for the chosen MCU board.
Display library	GxEPD2	Common and well-documented e-paper library for SPI panels.
Graphics library	Adafruit GFX	Used by GxEPD2 and easy to understand for simple screen layouts.

1. Read digital microphone samples at a fixed sample rate.
2. Apply A-weighting filter in firmware.
3. Compute RMS energy for the update window.
4. Convert the level to dBA with the stored calibration offset.
5. Update the display only when the visible state changes or on the scheduled refresh step.

SECTION 06

Display and User Interface

ELEMENT	RULE	PURPOSE
Main level field	Use the largest type on the screen.	The user must read the sound level from across the desk.
State field	Use one short word.	The user must know the condition at once.
Dose field	Show percent of daily target.	The user must see long-duration exposure.
Peak field	Show maximum recent value.	The user must see transient high sound.
Battery field	Show only if firmware battery readback is implemented.	Battery display is secondary to sound display in V1.

SCREEN STATE	TRIGGER	MEANING
NORMAL	Below 80 dBA	The room sound is within the normal target zone.
WARNING	80 dBA or more	The user should manage listening time or reduce level.
ALERT	85 dBA or more	The user should reduce level or shorten the session.
HIGH	94 dBA or more	The user should reduce level at once.

SIMPLICITY RULE

V1 does not need menus, graphs, Bluetooth setup, or animated transitions. The screen must behave like an instrument, not like an app.

SECTION 07 Bench Build Recipe

This is the compact beginner build order. Follow it exactly.

STEP	DO THIS	PASS RESULT
1	Connect the XIAO board to USB-C only. Confirm that the board enumerates and can run a trivial sketch.	Board powers correctly from USB-C.
2	Connect the LiPo battery. Confirm that the charge LED changes state when USB-C is attached and removed.	Battery charging path works.
3	Solder and wire the e-paper display. Load a full-screen black and full-screen white display test.	Display fully refreshes and is readable.
4	Load a microphone sample test. Print or display a raw level value. Clap near the board.	The value changes clearly with sound.
5	Combine display and microphone into one firmware build. Show current level, peak, and average.	The screen updates and the reading looks stable.
6	Use a reference meter and store one calibration offset.	The displayed level now tracks the reference well enough for personal use.

NO CASE YET

Do not start CAD or 3D printing until Step 6 passes. A failed bench prototype in a beautiful case is still a failed prototype.

DEFAULT FIRMWARE VALUES

Use these first: warning at 80 dBA, alert at 85 dBA, high state at 94 dBA, display update each second, one-minute rolling average, one-point calibration offset.

SECTION 08

Calibration and Accuracy

A digital microphone does not give true dBA by itself. The device needs calibration against a known reference. V1 should use a steady test sound and a reference meter at the same physical position.

The fixed calibration reference for this build is the Triplett SLM400 Sound Level Meter. It supports A weighting and fast or slow response, and it is sold as a Type 2 meter. Use A weighting and slow response during the one-point offset calibration.

1. Put the reference meter and the device at the same point.
2. Use steady pink noise or another stable test sound.
3. Wait for both readings to stabilize.
4. Read the two values.
5. Store the difference as the calibration offset.
6. Repeat the test after a major hardware change.

ACCURACY RULE

Do not claim legal meter accuracy. After calibration, use the value as a personal hearing-safety guide.

SECTION 09

Enclosure Design Recipe

The enclosure comes after the electronics pass the bench tests. The first enclosure should be easy to print and easy to reopen.

FEATURE	FINAL DECISION	WHY
Case style	Two-part wedge case	Simple to print and simple to service.
Front window	51 by 26 mm cutout	Large enough for the visible e-paper area with a small bezel.
Board mount	Thin foam tape for V1	Fast and forgiving for the first prototype.
Battery mount	Thin foam tape in a rear battery bay	No custom battery holder is needed in V1.
USB-C opening	Side cutout for the XIAO connector	Easy charging without opening the case.
Microphone port	One small hole aligned with the XIAO microphone area	Lets room sound reach the onboard microphone.

RECOMMENDED DIMENSION	VALUE	USE
Internal width	about 78 mm	Fits the display board width plus wire bend room.
Internal height	about 52 mm	Fits the display board and battery stack.
Internal depth	about 14 mm	Fits the battery, XIAO board, and wire slack.
Front angle	about 15 to 20 degrees	Makes the screen easy to read on a desk.

ENCLOSURE PHILOSOPHY

The first case should optimize serviceability, not elegance. Make it easy to open, easy to rewire, and easy to print again.

ENCLOSURE DETAIL	FINAL DECISION	COMMENT
Front bezel	About 1.2 mm visible lip around the window	Hides the board edge and keeps the screen opening clean.
Rear closure	Four printed screw bosses	Lets you reopen the case without destroying it.
USB-C cutout	Open-sided slot	Makes cable alignment easier for a first print.
Mic opening	Single 1.5 to 2.0 mm hole	Small enough to stay clean, large enough for room sound.

SECTION 10

Final Assembly

1. Print the case only after the bench build passes.
2. Fit the display board into the front half and confirm the viewing window alignment.
3. Route the eight display wires so they do not press on the screen surface.
4. Fix the XIAO board and battery into the rear half.
5. Check that the USB-C connector aligns with the side opening.
6. Close the case and power the device again.
7. Repeat the display test and microphone test.
8. Repeat the calibration after final assembly.

SECTION 11

Verification and Acceptance

TEST ID	TEST	EXPECTED RESULT
T-01	USB-C charge test	Battery charges and the device stays stable.
T-02	Battery-only run test	Device runs with no USB cable attached.
T-03	Display test	Full black, full white, and text screens all refresh correctly.
T-04	Microphone response test	Reading changes clearly with controlled sound changes.
T-05	Threshold test	State changes at the configured thresholds.
T-06	Calibration retention test	Offset remains after power cycle.

DONE MEANS THIS

The V1 build is done only when the device works on battery, shows readable values, agrees with the reference after calibration, and still works after it is put in the case.

SECTION A

Appendix A Shopping List

LINE	ITEM	QTY	BUY THIS
1	Seeed Studio XIAO nRF52840 Sense	1	Official Seeed product page for XIAO BLE Sense nRF52840.
2	Waveshare 2.13 inch e-Paper HAT, black and white	1	Official Waveshare 2.13 inch e-Paper HAT, 250 by 122, SPI.
3	Adafruit 3.7 V 2000 mAh Li-ion polymer battery	1	Adafruit product 2011, JST-PH, 60 by 36 by 7 mm.
4	28 AWG stranded hookup wire	1 pack	Use for the eight display wires.
5	Lead-free rosin-core solder	1 spool	Use standard electronic solder only.
6	USB-C cable	1	For power, programming, and charge.
7	Reference sound meter	1	Triplett SLM400 Sound Level Meter. Use this exact meter for V1 calibration work.